What one reference can establish after 111

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> For the specified independent gate model, a calibrated click probability on the ideal match state already gives a lower bound on the gate-path click probability. A separately observed ideal bit on each gate trial is unnecessary for this particular test.

The conditional state supplies the bound

Fix the earlier reported history w=111. Let P be the pure joint output of the known two-qubit unitary acting on its half of a fresh maximally entangled probe. Complete depolarization gives I/16 on the joint output. Under the supplied fresh selector, with unitary probability t=0.37 independent of w and of the apparatus, the conditional state just before the readout is

> $\rho_0 = tP+(1-t)I/16 = sP+(1-s)(I-P)/15$, $s=131/320$.

Let E be the binary click effect applicable at that history. Write $h=\mathrm{Tr}(EP)$ and $q=\mathrm{Tr}(E\rho_0)$. Positivity gives $q \geq sh$, since $\mathrm{Tr}[E(I-P)] \geq 0$. The effect $E=hP$ attains equality for every $0 \leq h \leq 1$. This statement allows an arbitrary quantum effect. It requires that the reference and gate trials at this history admit a common effective readout on the probe. Correlated apparatus side information about the current selector or route can violate that premise.

For the inherited target $q_\star = 81711338569043/224995905216000 \approx 0.36316811406$, a matched h strictly above $q_\star/s = 0.88712821756$ excludes exact agreement under this null. At the previously specified 99% reference response, the lower bound is 0.40528125, which exceeds the target by 0.04211314. These numbers refer to the conditional population; no gate or reference observations were supplied.

The operator condition is explicit. For any state ρ and any reference P, $q \geq \lambda h$ for every effect E if and only if $\rho - \lambda P$ is positive semidefinite. Sufficiency follows from the positive trace pairing. Necessity follows by taking E to project onto any negative eigenvector. Verifying only the scalar $\mathrm{Tr}(P\rho)$ does not certify that operator inequality.

A match probability leaves coherence unknown

Suppose the state is known only to satisfy $\mathrm{Tr}(P\rho)=s$, for the same rank-one reference P. Define $h=\mathrm{Tr}(EP)$ and $q=\mathrm{Tr}(E\rho)$. With the squared convention for fidelity, $F(P,\rho)=s$. Applying the measurement to both states can only increase fidelity, hence $\sqrt{s}\le\sqrt{(hq)}+\sqrt{((1-h)(1-q))}$. Solving the inequality gives the sharp lower envelope

> $q \ge [\max\{0, \sqrt{(sh)} - \sqrt{((1-s)(1-h))}\}]^2$.

For $s+h>1$, take the pure state with amplitudes $\sqrt{s}$ and $\sqrt{(1-s)}$ in a plane spanned by the reference and one orthogonal vector. Project onto the direction with amplitudes $\sqrt{h}$ and $-\sqrt{(1-h)}$. Squaring the overlap attains the bound. For $s+h\le1$, a scaled projector orthogonal to that state gives $q=0$ and reference response h. At $s=1$ the state is P and $q=h$ directly.

Using the same inherited s and $h=0.99$ now gives a minimum of 0.31333675. The target 0.36316811 is attainable: with the displayed projector, varying the state's off-diagonal entry between zero and its maximum magnitude continuously interpolates from 0.4111875 to 0.31333675. Thus an observed 99% reference response and a match population s, without the state structure on page 1, would allow the target. For $q\star<s$, strict exclusion on this information alone requires

> $h > [\sqrt{(s\, q\star)}+\sqrt{((1-s)(1-q\star))}]^2 = 0.99774815803\ldots$.

The distinction is a sensitivity check on the original state assumption. The depolarizing mixture on page 1 already has the stronger operator property. The unrestricted coherent state describes the uncertainty left if experimental characterization supports only the match population. The click-probability inequality is a consequence of standard quantum fidelity monotonicity.

Wilde, Quantum Information Theory, Ch. 9, fidelity under measurement.